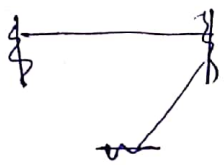
EE 354 - 2019

Question 1

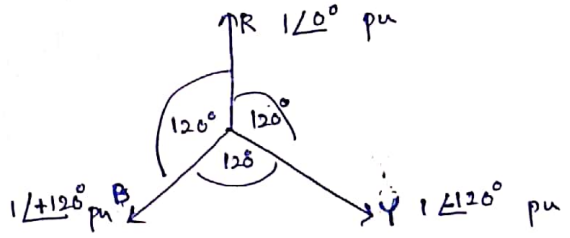


$V_R = 0.1\angle 0^\circ$ pu

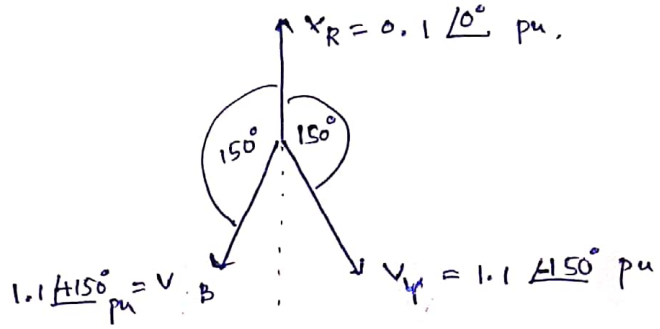
$V_Y = 1.1\angle -150^\circ$ pu

$V_B = 1.1\angle 150^\circ$ pu

1.1) Before the fault.



after the fault.



1.2)
$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda^2 & \lambda \\ 1 & \lambda & \lambda^2 \end{bmatrix} \quad (\lambda = 1\angle 120^\circ)$$

1.3) $X_1 = 0.1j$

$X_2 = 0.1j$

$X_0 = 0.05j$

$[V_{RYB}] = A[V_{R012}]$

$[V_{R012}] = \frac{1}{3}A^{-1}[V_{RYB}]$

$$\begin{bmatrix} V_{R0} \\ V_{R1} \\ V_{R2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda & \lambda^2 \\ 1 & \lambda^2 & \lambda \end{bmatrix}\begin{bmatrix} 0.1\angle 0^\circ \\ 1.1\angle -150^\circ \\ 1.1\angle +150^\circ \end{bmatrix}$$

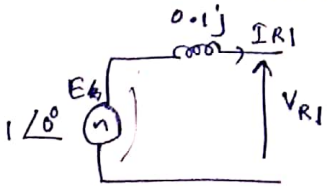
$V_{R0} = \frac{1}{3}(0.1\angle 0^\circ + 1.1\angle 150^\circ + 1.1\angle +150^\circ) = \frac{1.805}{3}\angle 180^\circ = -\frac{1.805}{3}$ pu

$= -0.602$ pu //

$V_{R1} = \frac{1}{3}(0.1\angle 0^\circ + 1\angle 120^\circ \times 1.1\angle 150^\circ + 1\angle 240 \times 1.1\angle +150^\circ) = 0.668$ pu //

$V_{R2} = \frac{1}{3}(0.1\angle 0^\circ + 1\angle 240 \times 1.1\angle 150^\circ + 1\angle 120^\circ \times 1.1\angle +150^\circ) = 0.033$ pu //

(+) seq. network



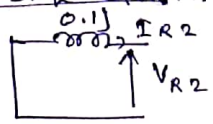
~~$I_{R1} = 1\angle 0$~~

$V_{R1} = E - 0.1j I_{R1}$

$I_{R1} = \frac{1\angle 0^\circ - 0.668}{0.1j} = 3.32\angle -90^\circ$

$= -3.32j$ pu //

(-) seq. network



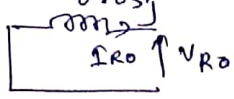
$V_{R2} = 0.1j I_{R2}$

$I_{R2} = \frac{0.033}{0.1j}$

$= 0.33\angle -90^\circ$

$= -0.33j$ pu

Zero seq. network



$V_{R0} = 0.05j \times I_{R0}$

$I_{R0} = \frac{-0.602}{0.05j}$

$= 12.04\angle 90^\circ$

$= 12.04j$ pu

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1.4) $V_{base} = 132\ kV$

$S_{base} = 100\ MVA$

$$I_{base} = \frac{S_{base}}{\sqrt{3}\ V_{base}} = \frac{100 \times 10^6}{\sqrt{3} \times 132 \times 10^3} = 437.3866\ A$$

$$[I_{RYB}] = A\,[I_{R012}]$$

$$\begin{bmatrix} I_R \\ I_Y \\ I_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \lambda^2 & \lambda \\ 1 & \lambda & \lambda^2 \end{bmatrix} \begin{bmatrix} I_{R0} \\ I_{R1} \\ I_{R2} \end{bmatrix}$$

$$I_R = I_{R0} + I_{R1} + I_{R2} = 12.04\angle 90^\circ + 3.32\angle -90^\circ + 0.33\angle -90^\circ = 8.39\angle 90^\circ\ pu$$

$$I_Y = I_{R0} + \lambda^2 I_{R1} + \lambda I_{R2} = 12.04\angle 90^\circ + 1\angle 240^\circ \times 3.32\angle -90^\circ + 1\angle 120^\circ \times 0.33\angle -90^\circ = 14.10\angle 100.58^\circ\ pu$$

$$I_B = I_{R0} + \lambda I_{R1} + \lambda^2 I_{R2} = 12.04\angle 90^\circ + 1\angle 120^\circ \times 3.32\angle -90^\circ + 1\angle 240^\circ \times 0.33\angle -90^\circ = 14.10\angle 79.42^\circ\ pu$$

$$I_R = 8.39\angle 90^\circ \times 437.3866 = 3669.67\angle 90^\circ\ A$$

$$I_Y = 14.10\angle 100.58 \times 437.3866 = 6167.151\angle 100.58^\circ\ A$$

$$I_B = 14.10\angle 79.42 \times 437.3866 = 6167.151\angle 79.42^\circ\ A$$

1.5) $\left.\begin{array}{l} V_R \neq V_Y \neq V_B \neq 0 \\ I_R \neq I_Y \neq I_B \neq 0 \end{array}\right\} \rightarrow$ ∴ not a ground fault. fault occurs in all 3 phases.

Since $I_{R0} \neq 0 \rightarrow$ unbalanced fault.

∴ This is a unbalanced LLL fault.

Fault currents:- $I_{RF} = 3.669\angle 90^\circ\ kA$

$I_{YF} = 6.167\angle 100.58^\circ\ kA$

$I_{BF} = 6.167\angle 79.42^\circ\ kA$

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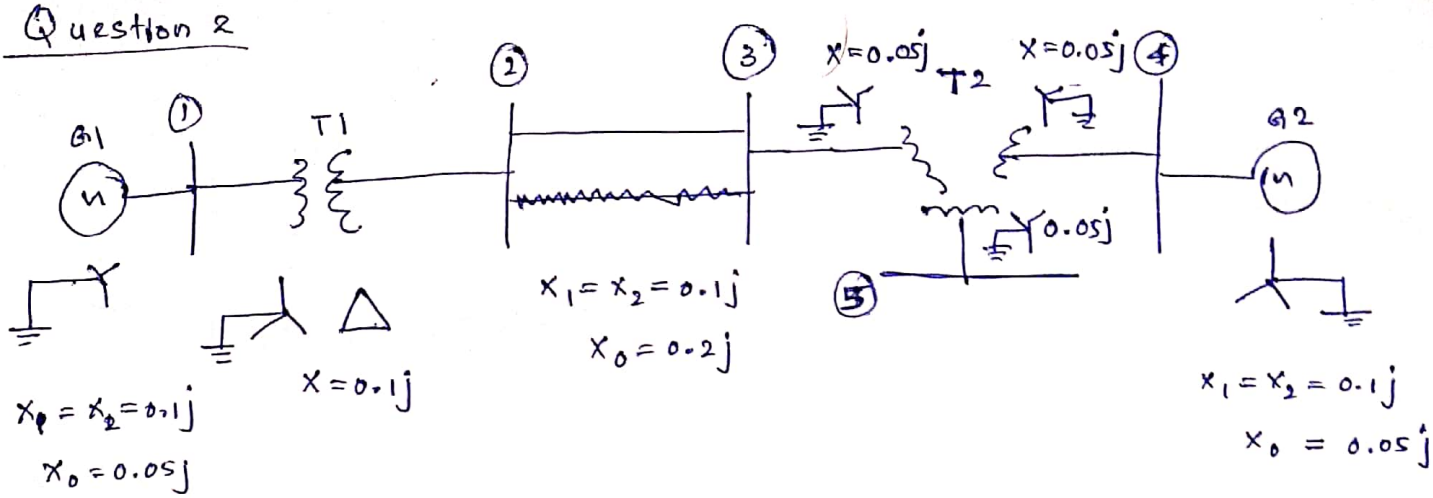
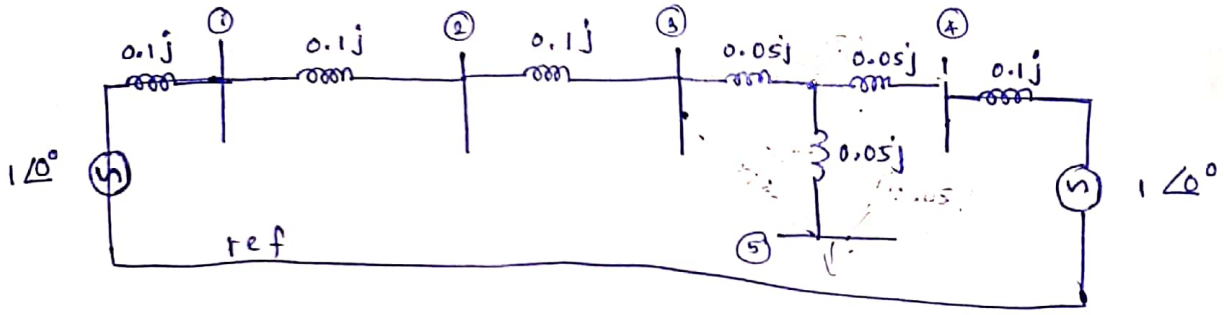
Question 2

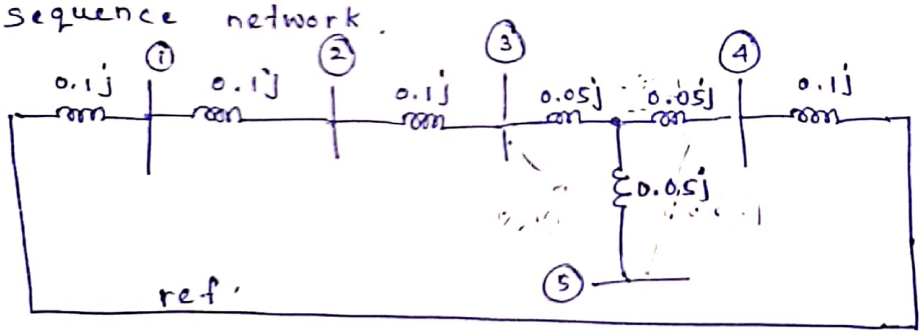
2.1)



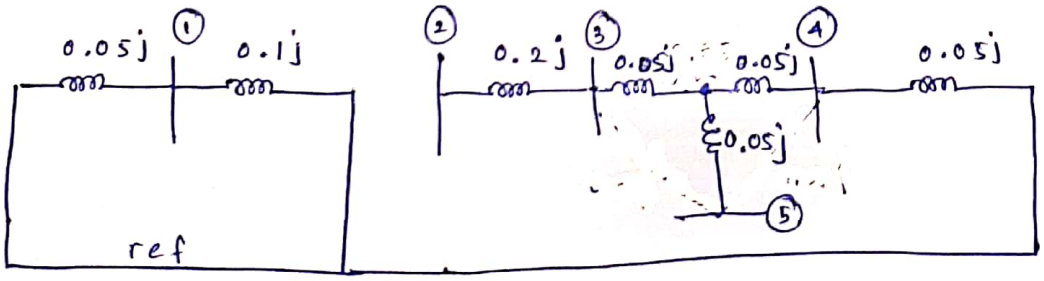
2.2) positive seq. network.



(-) sequence network.



zero sequence network.



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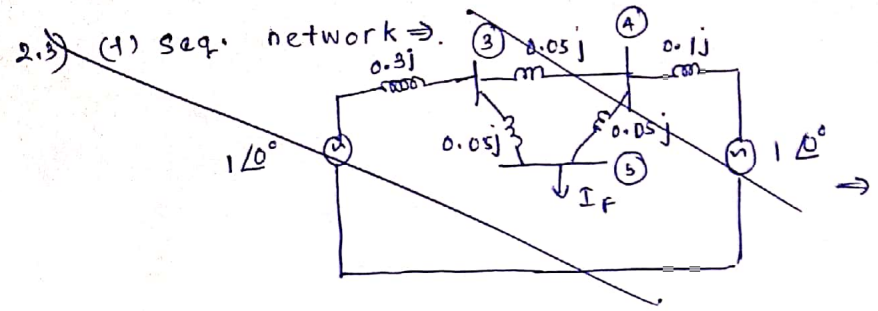
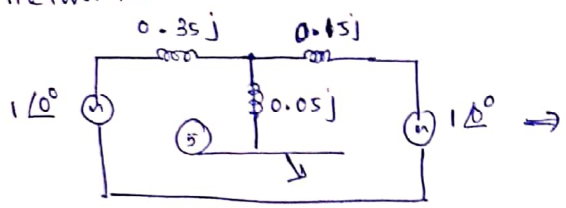
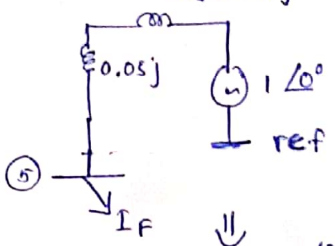
2.3) (+) seq. network ⇒.

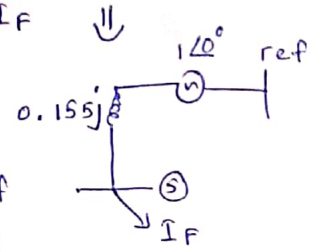


2.3) (+) seq. network.

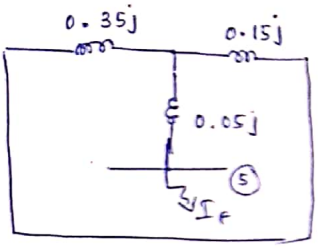


$0.35j \,//\, 0.15j = 0.105j$





(−) seq. network



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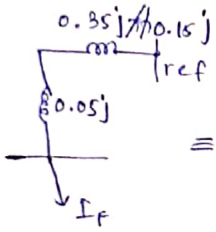

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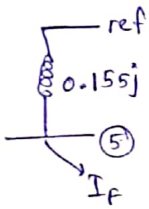
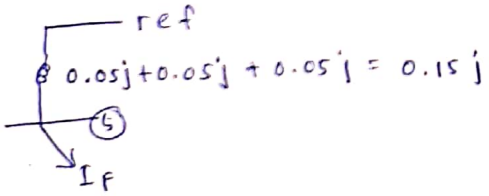
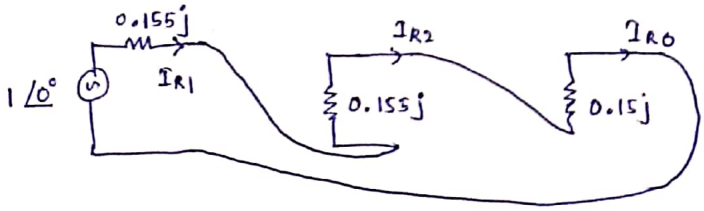


zero sequence network.



2.4) (a) SLG in R phase :- $V_R = 0$, $I_Y = I_B = 0$

connect seq. networks in series.

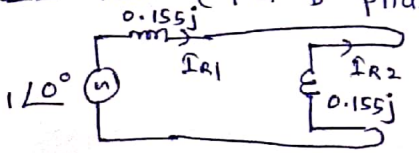


$I_{R1} = I_{R2} = I_{R0}$

$1\angle 0^\circ = I_{R1}(0.155j + 0.155j + 0.15j)$

$I_{R1} = 2.174 \angle -90^\circ$ pu //

(b) DL fault (Y & B phase) ; $I_R = 0$, $I_{YF} = -I_{BF}$



$I_{R1} = -I_{R2}$

$1\angle 0^\circ = I_{R1}(0.155j + 0.155j)$

$I_{R1} = 3.23 \angle -90^\circ$ pu //

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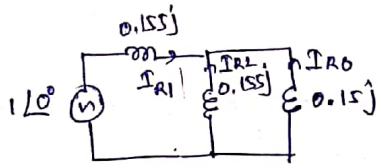
(c) DLG (Y & B phase) ; $I_F = I_{YF} + I_{BF}$

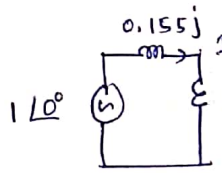
$V_Y = V_B = 0$



$I_{R1} + I_{R2} + I_{R0} = 0$

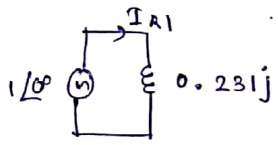
$V_{R1} = V_{R2} = V_{R0}$

|||



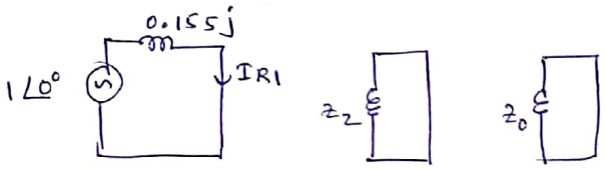
$0.155j \,//\, 0.15j = 0.076\angle 90° = 0.076j$

|||



$$I_{R1} = \frac{1\angle 0°}{0.231j} = 4.33\angle -90° \text{ pu} \;//$$

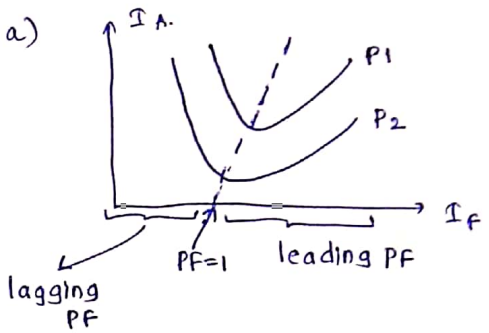
(d) LLL fault



$$I_{R1} = \frac{1\angle 0°}{0.155j} = 6.45\angle -90° \text{ pu} \;//$$

Question 3

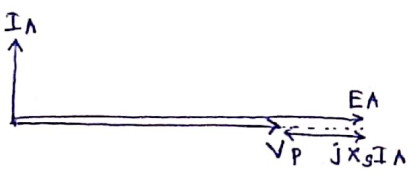
3.1) a)



$P_1 > P_2$

b) A synchronous motor with no rotor shaft visible to outside is known as a sync. condenser. i.e. it operates under no-load condition. Used to correct power factor in a power system. by operating it in over excited mode.

c)



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3.2) $V_{LL} = 230\,V = V_P$ $\quad$ Δ connected

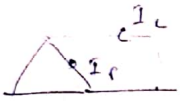
$S = 50\ kVA$ $\quad$ $f_e = 50\ Hz$

$\cos\theta = 0.85$ lagging $\quad$ Load = 32 hp = 23.8624 kW

$X_s = j2.75\ \Omega$

$R_A \rightarrow$ negligible

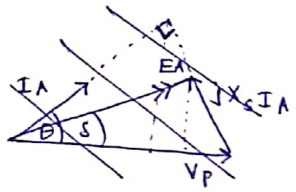
$P_{rot} = 3.5\ kW$



a) $E_A = ?$

operate at 0.82 PF lagging.

$\cos\theta = 0.82$



$P_{in} = \sqrt{3}\, V_{LL} I_{LL} \cos\theta$

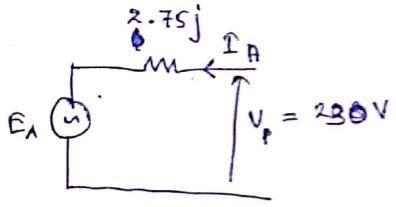
$P_{in} = P_{out} + P_{rot} + P$

$= 23.8624 + 3.5$

$= 27.3624\ kW$

$\therefore \sqrt{3}\, V_{LL} I_{LL} \cos\theta = 27.3624$

$I_L = \dfrac{27.3624 \times 10^3}{\sqrt{3} \times 230 \times 0.82} = 83.76\ A$



$I_A = |I_A| = \dfrac{I_L}{\sqrt{3}} = \dfrac{83.76}{\sqrt{3}} = 48.36\ A$

$\angle I_A = \theta = \cos^{-1} 0.82 = 34.92^\circ$

$\therefore I_A = 48.36 \angle 34.92^\circ\ A$

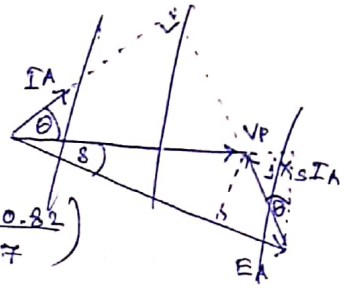
$E_A + jX_sI_A = V_P$

$E_A = 230 - j(2.75)(48.36\angle 34.92^\circ)$ ~~$= 347.487 \angle 12.655^\circ$~~ ~~V~~

~~$E_A = 188.59 \angle -144.675^\circ$ V~~

$E_A = 188.59 \angle -35.324^\circ$ V

method 2 :-



$\sin\delta = \dfrac{X_s I_A \cos\theta}{|E_A|}$

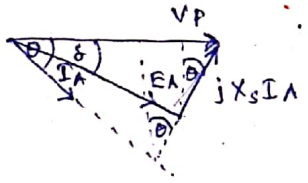
$\delta = \sin^{-1}\left(\dfrac{2.75 \times 48.36 \times 0.82}{188.597}\right)$

$\delta = 35.326^\circ$

~~$E_A^2 = (X_s I_A \cos\theta)^2 + (V_P + X_s I_A \sin\theta)^2$~~

~~$= (2.75 \times 48.36 \times 0.82)^2 + (230 + (2.75 \times 48.36 \times \sin(\theta)))^2$~~

~~$|E_A| = 133.85\ A$~~



$E_A^2 = (V_P - X_s I_A \sin\theta)^2 + (X_s I_A \cos\theta)^2$

$= (230 - (2.75 \times 48.36 \times \sin\theta))^2 + (2.75 \times 48.36 \times 0.82)^2$

$|E_A| = 188.597$

$E_A = 188.597 \angle -35.33^\circ$ V

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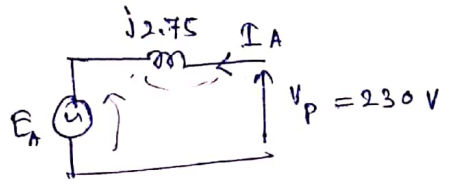
(b) $P_{out} = 40\ hp = 29.828\ kW$

$P_{in} = 29.828 + 3.5 = 33.328$

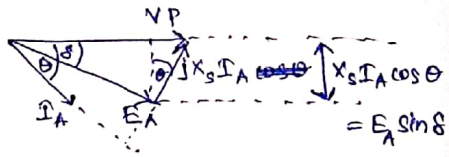
$|E_A|$ does not vary.



~~$|E_A| = 188.597 \angle -35.324^\circ$~~

~~$V_p = E_A + jX_s I_A$~~

~~$I_A = \dfrac{230 - 188.597 \angle -35.324^\circ}{j2.75} =$~~



$|E_A| = 188.597$

Using $P_{in} = \sqrt{3}\, V_p I_A \cos\theta = \left(\dfrac{3V_p}{X_s}\right)(X_s I_A \cos\theta) = \left(\dfrac{3V_p}{X_s}\right)(E_A \sin\delta)$.

$\sin\delta = \dfrac{33.328\times10^3 \times 2.75}{3\times230\times188.597} = 0.7043 \rightarrow \delta = 44.773$

$\therefore E_A = 188.597 \angle -44.773^\circ$

$V_p = E_A + jX_s I_A$

$I_A = 230 - 188.597 \angle -44.773^\circ / j\times2.75 = 59.62 \angle -35.89^\circ\ A$

(3.3) ✓

Question 4

4.1) ✓

4.2) a) transient unidirectional component

Steady state alternating component.

b) fault current $= (I'' - I')e^{-t/T''} + (I' - I_{ss})e^{-t/T'} + I_{ss}$

c) $S = 100\ MVA$, $V_{LL} = 13.5\ kV$, Y connected, 3ϕ, $F_c = 60\ Hz$

$X_s = 1\ pu$ $\quad T' = 1.15\ s$

$X' = 0.3\ pu$ $\quad T'' = 0.03\ s$

$X'' = 0.15\ pu$

~~$X'' = \frac{E_A}{I''} \rightarrow I'' = 0.15$~~

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$$X'' = \frac{E_A}{I''} \quad , \quad X' = \frac{E_A}{I'}$$

$$I''_{(pu)} = \frac{1\angle 0^\circ}{0.15} = \frac{20}{3} = 6.67 \text{ pu}$$

$$I'_{(pu)} = \frac{1\angle 0^\circ}{0.3} = \frac{10}{3} = 3.33 \text{ pu}$$

$$I_{base} = \frac{S_{base}}{\sqrt{3}\, V_{base}} = \frac{100 \times 10^6}{\sqrt{3} \times 13.5 \times 10^3} = 4.277 \text{ kA.}$$

$$\therefore I'' = 6.67 \times 4.277 = 28.53 \text{ kA}$$

$$I' = \cancel{6.67}\ 3.33 \times 4.277 = 14.24 \text{ kA.}$$

$$I_{ss\,(pu)} = \frac{E_A}{X_s} = \frac{1}{1 \text{ pu}} = 1 \text{ pu.}$$

$$I_{ss} = 1 \times 4.277 = 4.277 \text{ kA.}$$

$\therefore$ Fault current (rms)

$$= (28.53 - 14.24)\times 10^3\, e^{-t/0.03} + (14.24 - 4.277)\times 10^3\, e^{-t/1.15} + 4.277 \times 10^3$$

$$= 14290\, e^{-\frac{100}{3}t} + 9963\, e^{-\frac{20}{23}t} + 4.277 \times 10^3 \text{ A}$$

4.3)

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